FLUID MECHANICS

Subject Code: 23A01303

UNIT - 1: Basic Concepts and Definitions

B.Tech / JNTUA R23 / Handwritten Notes

Syllabus: Distinction between a fluid and a solid; Density, Specific weight, Specific gravity, Kinematic and dynamic viscosity; Variation of viscosity with temperature, Newton's law of viscosity; Vapor pressure, Boiling point, Surface tension, Capillarity, Bulk modulus of elasticity, Compressibility.

1. Introduction

Fluid Mechanics

It is the branch of science that deals with the behaviour of fluids (liquids and gases) at rest as well as in motion, and the forces acting on them. Its three branches are: Fluid Statics (fluids at rest), Fluid Kinematics (motion without considering forces) and Fluid Dynamics (motion with the forces causing it).

Fluid

A fluid is a substance that deforms continuously when even a very small shear force is applied to it, however small the force may be. Liquids and gases are fluids. Fluids have no definite shape; they take the shape of the container.

Distinction between a Solid and a Fluid

Property	Solid	Fluid
Shear force	Resists by a fixed deformation (angle) and then stops	Deforms continuously as long as the force acts
Shape	Definite shape and volume	No definite shape; takes shape of container
Molecular spacing	Very close, strong cohesive forces	Larger spacing, weak cohesive forces
Stress	Depends on strain (Hooke's law)	Depends on rate of strain (Newton's law)
Tensile strength	Can withstand tension, compression, shear	Can withstand only compression; almost no tension
Examples	Steel, wood, concrete	Water, oil, air, mercury

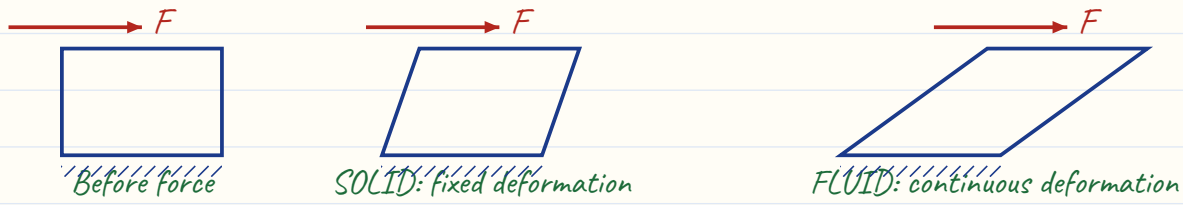


Fig 1.1: Response of a solid and a fluid to a shear force

Types of Fluids

- Ideal fluid – incompressible and has zero viscosity (imaginary, no shear stress). It is used for simplifying analysis.
- Real fluid – has viscosity (all actual fluids). Shear stress exists when there is relative motion.
- Newtonian fluid – obeys Newton's law of viscosity, $\tau = \mu (du/dy)$. Eg: water, air, kerosene, alcohol.
- Non-Newtonian fluid – does not obey the linear law. Eg: blood, paint, toothpaste, slurries, polymer solutions.
- Ideal plastic – shear stress exceeds a yield value, then remains constant for any strain rate.
- Compressible fluid – density changes with pressure (gases); Incompressible fluid – density is constant (liquids).

Continuum Concept

Fluid is made of molecules, but in engineering we assume it is a continuous medium (continuum): properties like density, pressure and velocity vary smoothly from point to point. This is valid when the characteristic length of the problem is much larger than the mean free path of molecules (Knudsen number is small).

Units and Dimensions (SI)

Quantity	Symbol	SI Unit	Dimension
Mass	m	kg	M
Length	L	m	L
Time	t	s	T
Force	F	$N = kg\cdot m/s^2$	$M L T^{-2}$
Pressure	p	$Pa = N/m^2$	$M L^{-1} T^{-2}$
Density	ρ	kg/m^3	$M L^{-3}$
Specific weight	w	N/m^3	$M L^{-2} T^{-2}$
Dynamic viscosity	μ	$N\cdot s/m^2 = Pa\cdot s$	$M L^{-1} T^{-1}$
Kinematic viscosity	ν	m^2/s	$L^2 T^{-1}$
Surface tension	σ	N/m	$M T^{-2}$
Bulk modulus	K	N/m^2	$M L^{-1} T^{-2}$

2. Basic Fluid Properties

2.1 Density (Mass Density), ρ

Mass of the fluid per unit volume.

$$\rho = m / V \ (kg/m^3)$$

- Water at 4°C: $\rho = 1000\ kg/m^3$; Sea water $\approx 1025\ kg/m^3$; Mercury $= 13600\ kg/m^3$; Air at 15°C, 1 atm $= 1.225\ kg/m^3$.
- Density of a liquid decreases slightly with temperature; for gases $\rho = p/(RT)$ (equation of state, $R = 287\ J/kg\cdot K$ for air).

2.2 Specific Weight (Weight Density), w or γ

Weight of the fluid per unit volume.

$$w = W / V = \rho\, g \ (N/m^3)$$

- Water: $w = 1000 \times 9.81 = 9810\ N/m^3$ ($9.81\ kN/m^3$). Air: $w = 1.225 \times 9.81 \approx 12.01\ N/m^3$.
- Unlike density, specific weight changes with location because g changes.

2.3 Specific Volume, v

Volume of fluid per unit mass. $v = V/m = 1/\rho$ (m^3/kg). Used mostly for gases.

2.4 Specific Gravity (Relative Density), S

Ratio of density (or specific weight) of the fluid to the density (specific weight) of a standard fluid. Standard fluid = water at 4°C for liquids; air (or hydrogen) at standard conditions for gases.

$$S = \rho_{fluid} / \rho_{water} = w_{fluid} / w_{water} \text{ (no units)}$$

- Mercury $S = 13.6$; Sea water ≈ 1.025; Petrol ≈ 0.72; Glycerin $= 1.26$; Kerosene ≈ 0.81.
- So $\rho = S \times 1000\ kg/m^3$ and $w = S \times 9810\ N/m^3$.

Example 1

A liquid of specific gravity 0.8 occupies 2 m^3. Find (a) density, (b) mass, (c) weight, (d) specific weight.

Sol: (a) $\rho = S \times 1000 = 0.8 \times 1000 = 800\ kg/m^3$

(b) $m = \rho V = 800 \times 2 = 1600\ kg$

(c) $W = mg = 1600 \times 9.81 = 15696\ N \approx 15.7\ kN$

(d) $w = \rho g = 800 \times 9.81 = 7848\ N/m^3$

Fluid (20°C)	ρ (kg/m^3)	S	μ (Pa·s)
Water	998	0.998	1.0×10^{-3}
Air	1.2	0.0012	1.8×10^{-5}
Mercury	13600	13.6	1.55×10^{-3}
Glycerin	1260	1.26	1.49
SAE 30 oil	917	0.917	0.29

3. Viscosity

Viscosity

It is the property of a fluid by virtue of which it offers resistance to the relative motion (shear deformation) between its adjacent layers. It is due to cohesion between molecules (liquids) and molecular momentum exchange (gases). It is the 'internal friction' of the fluid.

Newton's Law of Viscosity

Consider a fluid between two parallel plates, the lower plate fixed and the upper one moving with a velocity U under a force F. Fluid sticks to the plates (no-slip condition), so velocity varies from 0 at the fixed plate to U at the moving plate.

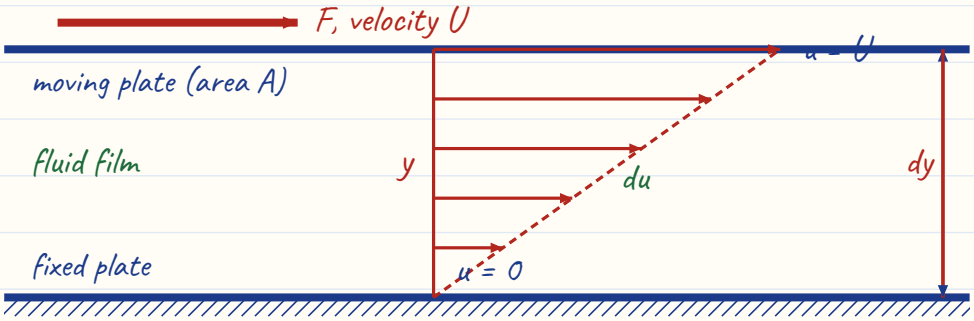


Fig 1.2: Velocity profile between two parallel plates

Newton's law states that the shear stress (τ) between adjacent fluid layers is directly proportional to the velocity gradient (rate of shear strain):

$$\tau = F/A = \mu \, (du/dy)$$

- τ = shear stress (N/m^2); du/dy = velocity gradient = rate of angular deformation (s^{-1}); μ = constant of proportionality = coefficient of dynamic viscosity.
- For linear profile: $du/dy = U/y$, so $\tau = \mu U / y$ and $F = \mu A U / y$.
- Fluids which obey this law are Newtonian fluids; the τ–du/dy graph is a straight line through the origin with slope μ.

Dynamic Viscosity (Absolute Viscosity), μ

$$\mu = \tau \, / \, (du/dy)$$

- SI unit: $N \cdot s/m^2 = Pa \cdot s = kg/(m \cdot s)$. CGS unit: poise ($dyne \cdot s/cm^2 = g/(cm \cdot s)$).
- 1 Pa·s = 10 poise; 1 poise = 0.1 $N \cdot s/m^2$; 1 centipoise = 0.001 Pa·s (water at 20°C ≈ 1 cP).
- Dimensions: $M L^{-1} T^{-1}$.

Kinematic Viscosity, ν

Ratio of dynamic viscosity to mass density. It is called 'kinematic' since force is not involved in its dimensions.

$$\nu = \mu / \rho$$

- SI unit: m^2/s. CGS unit: stoke (cm^2/s). 1 stoke = 10^{-4} m^2/s; 1 m^2/s = 10^4 stokes; 1 centistoke = 10^{-6} m^2/s.

Basic Concepts and Definitions

- Dimensions: $L^2 T^{-1}$. Water at 20°C: $\nu \approx 1 \times 10^{-6}$ m²/s (1 cSt); air: $\nu \approx 1.5 \times 10^{-5}$ m²/s.

Note: Water is 'thicker' (larger μ) than air, but air has a larger kinematic viscosity because its density is very small.

Types of Fluids on the Basis of Newton's Law

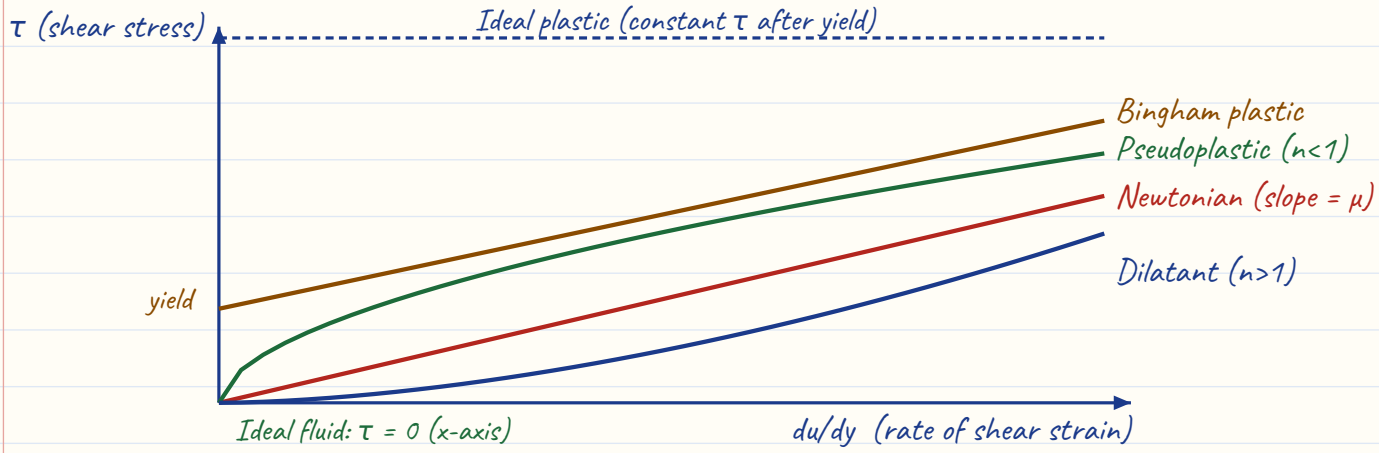


Fig 1.3: Rheological diagram (shear stress vs rate of shear strain)

- Newtonian: $\tau = \mu (du/dy)$. Constant μ. Water, air, oils, gasoline.
- Non-Newtonian: $\tau = A (du/dy)^n + B$. $n < 1$ → pseudoplastic (blood, paint, polymer solution; apparent viscosity falls as shear increases); $n > 1$ → dilatant (quicksand, starch suspension; viscosity rises with shear).
- Bingham plastic: needs a minimum yield stress before it flows, then behaves linearly (toothpaste, sewage sludge, mud).
- Ideal fluid: $\tau = 0$ for any du/dy. Elastic solid: $du/dy = 0$ for any τ (the y-axis).

Variation of Viscosity with Temperature

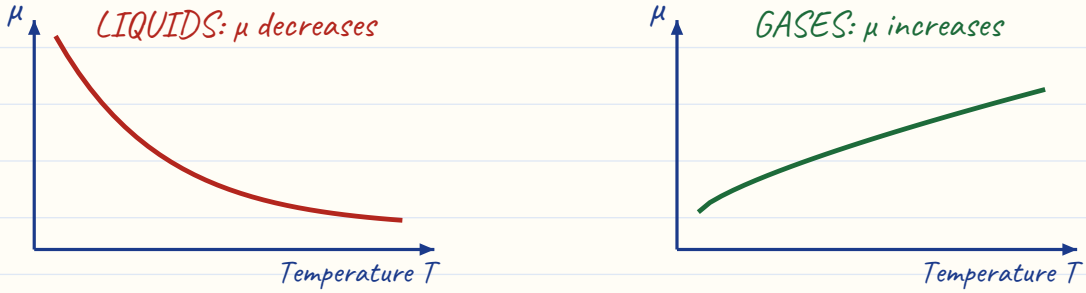


Fig 1.4: Effect of temperature on viscosity

- Liquids: viscosity decreases as temperature increases. Reason: viscosity is due to cohesive forces between molecules. On heating, molecules gain energy and cohesion weakens. (Engine oil flows easily when hot.)
- Gases: viscosity increases as temperature increases. Reason: viscosity is due to molecular momentum transfer between layers. On heating, molecular agitation rises, the exchange of momentum increases and so does the resistance.

Basic Concepts and Definitions

- Liquids – Andrade's equation: $\mu = A\, e^{B/T}$ or $\mu = \mu_0 / (1 + \alpha t + \beta t^2)$ (t in °C; α, β are constants).
- Gases – Sutherland's law: $\mu = C\, T^{3/2} / (T + S)$ (S = Sutherland constant, ≈ 110.4 K for air).
- Pressure has negligible effect on viscosity except at very high pressures.

Solved Problems on Viscosity

Example 2 – Flat plate over a fixed plate

The space between two parallel plates, 2.5 mm apart, is filled with oil of viscosity 0.9 N·s/m². The upper plate moves at 1.5 m/s. Find the shear stress in the oil. If plate area is 0.4 m², find the force needed.

Sol: $du/dy = U/y = 1.5 / 0.0025 = 600\ s^{-1}$

$\tau = \mu\, du/dy = 0.9 \times 600 = 540\ N/m^2$

$F = \tau\, A = 540 \times 0.4 = 216\ N$

Example 3 – Unit conversion

An oil of specific gravity 0.9 has a dynamic viscosity of 6 poise. Find its kinematic viscosity in stokes.

Sol: $\mu = 6$ poise $= 6 \times 0.1 = 0.6\ N·s/m^2$; $\rho = 0.9 \times 1000 = 900\ kg/m^3$

$\nu = \mu/\rho = 0.6/900 = 6.667 \times 10^{-4}\ m^2/s = 6.667 \times 10^{-4} \times 10\text{^}4 = 6.667$ stokes

Example 4 – Block sliding on an inclined plane

A square block (0.3 m side) weighing 200 N slides down a 20° incline on an oil film of thickness 0.2 mm and $\mu = 0.1\ N·s/m^2$. Find the steady velocity.

Sol: Driving force along plane $= W \sin 20° = 200 \times 0.3420 = 68.4\ N$

At steady speed the viscous force equals this force: $\tau = F/A = 68.4 / (0.3 \times 0.3) = 760\ N/m^2$

$\tau = \mu\, U / y \rightarrow U = \tau\, y / \mu = 760 \times 0.0002 / 0.1 = 1.52\ m/s$

Example 5 – Journal bearing (torque and power)

A shaft of diameter 80 mm rotates at 300 rpm in a bearing of length 60 mm. Radial clearance = 0.2 mm filled with oil, μ = 0.1 Pa·s. Find torque and power lost.

Sol: $U = \pi DN/60 = \pi \times 0.08 \times 300/60 = 1.2566$ m/s

$\tau = \mu U / c = 0.1 \times 1.2566 / 0.0002 = 628.3$ N/m^2

Area $A = \pi D L = \pi \times 0.08 \times 0.06 = 0.01508$ m^2; $F = \tau A = 9.475$ N

Torque $T = F \times D/2 = 9.475 \times 0.04 = 0.379$ N·m

Power $P = T\omega = 0.379 \times (2\pi \times 300/60 = 31.42) = 11.9$ W

4. Vapor Pressure, Boiling Point and Cavitation

Vapor Pressure (p_v)

When a liquid is in a closed vessel, molecules escape from the surface into the space above and exert a pressure on it. At equilibrium (rate of evaporation = rate of condensation) this pressure is the saturation vapor pressure. It increases with temperature.

- Water: $p_v \approx 2.34$ kPa at 20°C; 7.4 kPa at 40°C; 101.3 kPa at 100°C.
- Liquids with high vapor pressure (petrol, alcohol) are called volatile.

Boiling Point

A liquid boils when its vapor pressure becomes equal to the surrounding (absolute) pressure. Hence boiling point decreases when the external pressure is reduced (water boils below 100°C on a mountain; in a pressure cooker boiling point rises above 100°C).

Cavitation

If, in a flowing liquid, the local absolute pressure falls to the vapor pressure, the liquid vaporises and vapor bubbles form. When these bubbles are carried to a high pressure zone they collapse suddenly (implode), producing very high local pressures (hundreds of MPa), noise, vibration, pitting and erosion of metal surfaces, and fall in efficiency.

- Occurs in pump suction, turbine runner blades, propellers, venturi throats, and control valves.
- Prevention: keep pressure above p_v (reduce suction lift, increase NPSH), use smooth surfaces, cavitation-resistant materials, lower speeds.

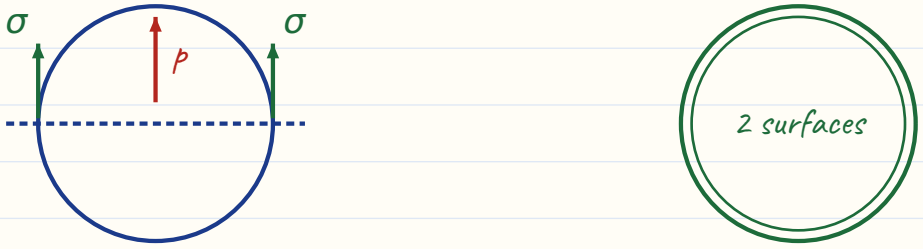
5. Surface Tension

Surface Tension (σ)

Surface tension is the tensile force acting on the surface of a liquid in contact with a gas (or another immiscible liquid) that tends to minimise the surface area. It is the force per unit length of a line drawn on the surface, perpendicular to it. Unit: N/m. Dimension: $M T^{-2}$.

- Cause: Cohesion. A molecule inside the liquid is pulled equally in all directions. A molecule at the surface has a net inward pull, so the surface behaves like a stretched elastic membrane.
- Examples: small droplets are spherical, a needle floats on water, insects walk on water, formation of soap bubbles.
- σ decreases as temperature rises; soaps and detergents reduce σ.
- Values at 20°C: water–air = 0.0728 N/m (0.073); mercury–air = 0.465–0.52 N/m; alcohol–air ≈ 0.022 N/m; soap solution ≈ 0.025 N/m.

Pressure Inside a Droplet, Soap Bubble and Liquid Jet



Droplet: $\Delta p = 4\sigma/d$ Soap bubble: $\Delta p = 8\sigma/d$

Fig 1.5: Forces on a half droplet / bubble

(i) Hollow droplet of diameter d (one free surface). Consider half the droplet. Pressure force (excess pressure Δp over projected area) balances the surface tension force around the circumference:

$$\Delta p \times (\pi/4)\, d^2 = \sigma \times \pi\, d \rightarrow \Delta p = 4\sigma / d$$

(ii) Soap bubble has two surfaces (inner and outer film), so the tension force is doubled:

$$\Delta p \times (\pi/4)\, d^2 = 2 \times \sigma\, \pi\, d \rightarrow \Delta p = 8\sigma / d$$

(iii) Liquid jet of diameter d and length L: $\Delta p \times (d\, L) = 2\sigma\, L \rightarrow \Delta p = 2\sigma / d$

Note: Smaller the droplet, larger the internal pressure. Δp is the excess of internal pressure over the outside pressure (gauge).

Example 6
Find the excess pressure inside (a) a water droplet of diameter 0.5 mm (σ = 0.0736 N/m) and (b) a soap bubble of 40 mm diameter (σ = 0.025 N/m).
Sol: (a) $\Delta p = 4\sigma/d = 4 \times 0.0736 / 0.0005 = 588.8\ N/m^2$
(b) $\Delta p = 8\sigma/d = 8 \times 0.025 / 0.04 = 5\ N/m^2$

6. Capillarity

Capillarity
The phenomenon of rise or fall of a liquid in a small-diameter tube (capillary) relative to the general level of the liquid outside is called capillarity. It occurs due to the combined effect of cohesion (liquid–liquid attraction) and adhesion (liquid–solid attraction).

- If adhesion > cohesion (water–glass), the liquid wets the wall, meniscus is concave, contact angle $\theta < 90°$ → capillary rise.
- If cohesion > adhesion (mercury–glass), meniscus is convex, $\theta > 90°$ ($\approx 130°$–$140°$) → capillary depression.



Fig 1.6: Capillary rise (water) and depression (mercury)

Expression for Capillary Rise / Fall

Let a tube of diameter d be dipped in a liquid of density ρ and surface tension σ, with contact angle θ. Let h be the height of rise.

- Upward force due to surface tension = $\sigma \cos\theta \times$ (circumference) = $\sigma \cos\theta \times \pi d$
- Weight of liquid column raised = $\rho g \times (\pi/4) d^2 \times h$

Equating them (for equilibrium):

$$\sigma \cos\theta\ \pi d = \rho g (\pi/4) d^2 h$$
$$h = 4\sigma \cos\theta / (\rho g d)$$

- For water–clean glass $\theta \approx 0°$, so $h = 4\sigma/(\rho g d)$. For mercury $\theta \approx 130°$, $\cos\theta$ negative → h negative = depression.

- $h \propto 1/d$ – narrower the tube, greater the rise. Capillarity is negligible for $d > 12$ mm.
- Applications: rise of water in soil and plants, ink in blotting paper, oil in a lamp wick. Piezometers/manometer tubes should be > 12 mm bore to avoid capillary errors.

Example 7

Calculate the capillary rise of water ($\sigma = 0.0736$ N/m, $\theta = 0°$) in a 2 mm glass tube. Also find the depression of mercury ($\sigma = 0.52$ N/m, $\theta = 130°$, $\rho = 13600$ kg/m^3) in the same tube.

Sol: Water: $h = 4\sigma \cos\theta / (\rho g d) = 4 \times 0.0736 \times 1 / (1000 \times 9.81 \times 0.002) = 0.015$ m = 15 mm (rise)

Mercury: $h = 4 \times 0.52 \times \cos 130° / (13600 \times 9.81 \times 0.002) = (2.08 \times -0.6428)/266.8 = -0.00501$ m

→ depression = 5.0 mm

7. Compressibility and Bulk Modulus of Elasticity

Compressibility (β)

Compressibility is the fractional change in volume per unit increase in pressure. It is the reciprocal of bulk modulus. $\beta = 1/K$ (m^2/N or Pa^{-1}).

Bulk Modulus of Elasticity (K)

Ratio of the increase in pressure to the corresponding volumetric strain. It measures resistance of the fluid to compression.

$$K = -dp / (dV / V) = dp / (d\rho / \rho)$$
$$\beta = 1/K$$

- The negative sign shows that volume decreases when pressure increases (dV is negative, so K is positive).
- Unit of K: N/m^2 (Pa). Dimension $M L^{-1} T^{-2}$. Water: $K \approx 2.1 - 2.2$ GPa (2.1×10^9 N/m^2); Mercury ≈ 26 GPa; Oil ≈ 1.5 GPa; Air (at 1 atm, isothermal) = 101 kPa.
- Large K → fluid is nearly incompressible (liquids). Small K → highly compressible (gases).
- For a perfect gas: isothermal process $K = p$; adiabatic process $K = k\,p$ ($k = cp/cv = 1.4$ for air).
- Speed of sound in a fluid: $c = \sqrt{(K/\rho)}$. Water: $c = \sqrt{(2.2 \times 10^9/1000)} \approx 1483$ m/s; air at 15 °C: $c = \sqrt{(kRT)} \approx 340$ m/s.

- Liquids are treated as incompressible in most problems, except water hammer, deep-ocean problems and acoustics.

Example 8

A liquid compressed in a cylinder has a volume of 0.0125 m^3 at 80 N/cm^2 and 0.0124 m^3 at 150 N/cm^2. Find the bulk modulus and compressibility.

Sol: $\Delta p = 150 - 80 = 70\ N/cm^2 = 70 \times 10^4 = 7 \times 10^5\ N/m^2$

$\Delta V = 0.0124 - 0.0125 = -0.0001\ m^3$; $\Delta V/V = -0.0001/0.0125 = -0.008$

$K = -\Delta p/(\Delta V/V) = 7 \times 10^5 / 0.008 = 8.75 \times 10^7\ N/m^2$

$\beta = 1/K = 1.143 \times 10^{-8}\ m^2/N$

Example 9

Water has $K = 2.1 \times 10^9\ N/m^2$. By how much (percent) does its volume decrease when the pressure is raised by 5 MPa?

Sol: $|\Delta V|/V = \Delta p/K = 5 \times 10^6 / 2.1 \times 10^9 = 2.38 \times 10^{-3} = 0.238\ \%$ (so water is practically incompressible)

8. Quick Revision Sheet

Property	Formula	SI Unit
Density	$\rho = m/V$	kg/m^3
Specific weight	$w = \rho g$	N/m^3
Specific gravity	$S = \rho/\rho_{water}$	–
Shear stress	$\tau = \mu\, du/dy$	N/m^2
Dynamic viscosity	$\mu = \tau/(du/dy)$	Pa·s
Kinematic viscosity	$\nu = \mu/\rho$	m^2/s
Droplet pressure	$\Delta p = 4\sigma/d$	N/m^2
Soap bubble	$\Delta p = 8\sigma/d$	N/m^2
Liquid jet	$\Delta p = 2\sigma/d$	N/m^2
Capillary rise	$h = 4\sigma \cos\theta/(\rho g d)$	m
Bulk modulus	$K = -dp/(dV/V)$	N/m^2
Compressibility	$\beta = 1/K$	m^2/N

Important Short Questions

1. Define fluid. How does it differ from a solid?
2. State Newton's law of viscosity. What are Newtonian and non-Newtonian fluids?
3. Why does viscosity of liquids fall and that of gases rise with temperature?
4. Differentiate dynamic and kinematic viscosity; give their units and conversions.
5. What is cavitation? Mention its effects and remedies.
6. Derive the expression for capillary rise in a tube. Why does mercury show depression?
7. Derive the pressure inside a droplet and a soap bubble.
8. Define bulk modulus and compressibility. What is their relation?

Conversion Cheat-Sheet

- 1 poise = 0.1 Pa·s; 1 cP = 10^{-3} Pa·s; 1 stoke = 10^{-4} m^2/s; 1 cSt = 10^{-6} m^2/s
- 1 kgf = 9.81 N; 1 bar = 10^5 Pa; 1 atm = 101.325 kPa = 760 mm Hg; g = 9.81 m/s^2